

Contractile rate of force development after anterior cruciate ligament reconstruction—a comprehensive review and meta-analysis

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Study design: Comprehensive review and meta-analysis.

Background: The recovery in rapid force production measured as the rate of force development (RFD) is not clear after anterior cruciate ligament reconstruction (ACLR).

Objectives: To evaluate (a) time-course change of between-limb asymmetries in isometric knee extension/flexion RFD in individuals post-ACLR and (b) differences in RFD between individuals post-ACLR and healthy controls.

Methods: A literature search of Web of Science, SPORTDiscus, PubMed-MEDLINE, and ScienceDirect identified 10 eligible studies (n = 246) assessing RFD after ACLR.

Results: Standard mean difference (SMD) for early-phase (<100 ms) knee extensor RFD was -1.07 (95% CI: $-1.46, -0.68$) when comparing ACLR vs uninjured limb, while SMD for late-phase (≥ 100 ms) RFD was -0.85 (95% CI: $-1.27, -0.42$). SMD for early- and late-phase knee flexor RFD was -0.74 (95% CI: $-1.19, -0.29$) and -0.79 (95% CI: $-1.19, -0.39$), respectively. Comparing ACLR limbs to uninjured controls, knee extensor SMD for early- and late-phase RFD was -1.42 (95% CI: $-2.10, -0.73$) and 1.09 (95% CI: $-1.81, -0.38$). For the knee flexors, SMD for early- and late-phase RFD was -0.78 (95% CI: $-1.96, -0.39$) and -1.14 (95% CI: $-1.60, -0.67$).

Conclusions: Anterior cruciate ligament reconstruction limbs demonstrated sustained post-surgical suppression in RFD capacity for the knee extensors/flexors compared to the contralateral limb as well as to healthy controls. Monitoring of RFD should be considered throughout rehabilitation and return to sport (RTS) after ACLR to assess the effectiveness of post-operative rehabilitation. Post-surgical ACLR rehabilitation should include training interventions to enhance RFD.

KEYWORDS

ACLR reconstruction, knee extensors, knee flexors, rate of force development, return to sport

1 | INTRODUCTION

Anterior cruciate ligament (ACL) rupture is a common knee ligament injury in various athletic populations.¹ Following ACL rupture, ACL reconstruction (ACLR) is often suggested to athletes who wish to return to their former level of sport performance.²⁻⁴ The goal of ACLR is to restore mechanical knee stability^{5,6} and regain functional capacity of the affected limb compared to that of the contralateral (uninjured) limb to facilitate a return to the pre-injury performance level.^{2,7} Nevertheless, ACLR itself and the post-surgical tissue response, including persistent knee effusion,⁸⁻¹⁰ cause functional deficits in knee range of motion.¹¹⁻¹³ Additionally, arthrogenic muscle inhibition (AMI)¹⁴⁻¹⁶ and gamma-loop dysfunction¹⁷ may cause deficits in muscle strength.¹⁸⁻²⁰ Adequate rehabilitation is essential when the athlete's goal is to return to competitive sport participation that includes maximal jumping and cutting movements.²¹ Specifically, the goal of rehabilitation after ACLR is to regain anterior knee stability and range of motion as well as to restore mechanical thigh muscle function,²² in order to create a functionally stable knee⁵ and ensure a safe and timely return to sport (RTS).²³ Consequently, resistance training after ACLR is often prescribed as an integral part of the rehabilitation process²⁴ in order to restore neuromuscular function, counter muscle atrophy, and retrain thigh muscle strength to approach that of the uninjured limb.²⁵

Regaining quadriceps and hamstring muscle strength is a common objective criterion for RTS.²⁶ Muscle strength is typically compared to the uninjured limb using a limb symmetry index (LSI) and measured on a dynamometer to obtain dynamic (isokinetic) or isometric strength values.^{25,27} LSI criteria for RTS vary between studies with recommendations ranging from 70% to 90%.^{25,28,29} Calculation of the hamstring-to-quadriceps (H/Q) strength ratio (ratio of knee flexor to knee extensor strength) also has been used to assess the capacity for dynamic knee joint stabilization provided by the quadriceps and hamstring muscles to monitor modifiable ACL injury risk factors.^{27,30-33}

Despite intensive rehabilitation and resistance training, thigh muscle strength may not be fully restored following ACLR since weakness of the knee extensors and flexors may persist for years following knee injury.^{19,20,34} In addition, associations between various RTS outcomes and maximal knee extensor strength have been indicated previously.^{26,35} Notably, a positive relationship between quadriceps strength asymmetry and ACL re-injury rate has been observed, implicating that for every 1% decrease in strength asymmetry, the incidence of re-injury may be reduced by 3%.³⁵ Among young active ACLR individuals, less than half may fulfill a $\geq 90\%$ LSI criterion for quadriceps strength.^{7,36,37} Within ACL-injured adults, Welling et al³⁸ reported that only 13 patients out of 62 (21%) reached $\geq 90\%$ LSI criterion for quadriceps

strength 9 months after ACLR. In a study by Cristiani et al⁸ where 4093 patients were assessed 6 months after ACLR, 36% and 47% reached the $\geq 90\%$ LSI criterion for quadriceps and hamstring strength, respectively.

However, the relationship between the ability to generate muscle force rapidly defined as the rate of force development (RFD) and various specific RTS outcomes is unexplored and hence remains unclear.²³ RFD reflects the ability to produce a high rate of rise in muscle force per unit of time following contraction onset, which can be measured during isometric contraction conditions using force-time or torque-time curves and calculating the average tangential slope between given time points on the curve (eg, at 0-50 ms).³⁹ In typical activities of daily living (eg, descending stairs, walking fast, or reversing a fall) and athletic movements (eg, sprinting, kicking, and jumping), short-duration muscle contraction times of 50-200 ms are often observed.^{39,40} However, when obtained at standardized isometric and optimal length-tension conditions maximal voluntary contraction (MVC) force is not reached until 300-500 ms from the onset of the contraction.⁴¹ Notably also, the time to reach joint angle and angular velocity-specific MVC torque is shorter for concentric vs eccentric and isometric contractions and decreases as a function of contractile speed.^{42,43} Consequently, with trained athletes and non-athletes alike, it becomes essential to assess RFD.^{39,40,44}

Physiological (ie, neural factors and peripheral muscular factors) determinants of contractile RFD have a distinct but also overlapping role in the early (<100 ms) and late phase (≥ 100 ms) of rising muscle force.³⁹ Motor unit recruitment and motor neuron discharge rate have a marked influence on the expression of early-phase volitional RFD (<75 ms), and intrinsic muscle properties, such as type II muscle fiber distribution, may also explain inter-individual and inter-muscular differences in this phase.⁴⁰ In contrast, quantitative muscle properties such as CSA (or volume) and MVC may have a stronger influence on late-phase RFD (≥ 100 ms).^{39,40} Specifically, in an uninjured population, late-phase RFD is governed not only by maximal muscle strength but also by the magnitude of efferent neural drive and muscle architecture properties such as fascicle pennation angle.^{39,40} It is important to note that the specific analysis techniques affect the interpretation of RFD data. Assessing RFD tangentially, using average slope analysis from the onset of muscle contraction (eg, 0-100 ms), reflects the combined influence of neural and muscular properties for the time interval in question. However, when assessing sequential RFD in an already activated muscle (eg, at 100-150 ms),^{45,46} neural activation may be already high/maximal, suggesting a dependency on peripheral physiological determinants of late-phase RFD.³⁹ This indicates within uninjured people that sequential RFD may be valuable when evaluating the relative contribution of neural and intrinsic muscle contractile factors on the expression of RFD during the early and late phases of rising

contraction force.^{39,40} However, while these data were obtained exclusively in healthy individuals, it remains uncertain if similar relationships would exist in injured populations, where neural activation might be slowed at the start of contraction and its relevance still be important for late-phase RFD.

Knee extensor RFD assessed with isokinetic and isometric dynamometry has been an outcome measure of interest following different types of knee surgery, such as total knee arthroplasty (TKA)^{47,48} and arthroscopic partial meniscectomy (APM)^{49,50} (Table S1). However, an uncoupling between efferent neural drive (active state rise) and mechanical muscle force output may be present when calculating RFD during isokinetic or dynamic (anisometric) test conditions (ie, force measurements interact with changes in joint angle and velocity in a non-linear way, in turn confounding the RFD measurement).⁵¹ Therefore, RFD test-retest reliability may be low in dynamic test settings⁵¹ in particular for the assessment of early-phase RFD.⁴⁰ Consequently, while RFD quantified during dynamic contractions may be more specific to athletic movement patterns,^{39,52,53} isolated isometric single-joint assessments appear to provide a more controlled experimental setting for RFD testing.^{39,40} Finally, when considering available assessment methods/devices for clinicians working with patients/athletes, static (isometric) assessment of RFD may be more feasible, affordable, and reproducible to perform compared to dynamic assessments, for example, using isokinetic dynamometry.

In order to evaluate the utility of quadriceps and hamstring muscle RFD testing during the time course of rehabilitation following ACLR, the aim of the present meta-analysis and systematic review was (a) to describe within-subject between-limb asymmetries in RFD and (b) compare RFD between individuals with and without ACLR during isometric knee extension and knee flexion assessments.

2 | METHODS

2.1 | Study design

This study adhered to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)⁵⁴ guidelines.

2.2 | Eligibility criteria

Only studies published in international English-language peer-reviewed journals were included. As no RCT studies could be identified, non-randomized observational group

studies were included in the analysis. Studies were included if (a) the study population had sustained unilateral ACL injury and subsequent ACLR, (b) the activity status of the study population was informed using the Tegner activity scale ($\geq 5/10$)⁵⁵ or to be at least recreationally active, and/or (c) comprising isometric knee extension or flexion RFD assessments in individuals post-ACLR (ACLR vs uninjured limb) and/or additionally compared to knee extensor/flexor RFD data obtained in healthy uninjured controls.

2.3 | Information sources and literature search

A comprehensive literature search was conducted in August 2019. Electronic databases comprised Web of Science, SPORTDiscus, PubMed-MEDLINE, and ScienceDirect. Other sources were not used. Search keywords were “anterior cruciate ligament,” “ACL,” “ACL-R,” “ACL injury,” “ACL rupture,” and “ACL reconstruction.” Keywords were combined with the Boolean operator “AND” independently with “rate of torque development,” “rate of force development,” “RTD,” “RFD,” or “explosive force” (see Appendices S1 and S2).

2.4 | Data extraction

Main outcome variables RFD or RTD were independently extracted from all studies included in the analysis. Descriptive statistics from a study by Mirkov et al⁵⁶ were retrieved by personal communication since they were not provided in the original publication. Also, the post-operative time period, ACLR graft type, and assessment protocol (knee flexion angle) were extracted. Further, study characteristics such as sample size, ratio of male to female participants, age, height, weight, and activity status or Tegner activity score⁵⁵ were extracted from ACLR groups and also from healthy control groups whenever possible. Descriptive information (mean, standard deviation, and percentage) and inferential statistical information (*P* values) were collected when available, to (a) describe the magnitude of between-limb asymmetry in individuals post-ACLR and (b) compare individuals post-ACLR to healthy controls.

2.5 | Study quality

Study quality was assessed using the Newcastle-Ottawa Scale (NOS) (Table S2), which evaluates three main aspects: selection of the study groups, comparability of the groups, and ascertainment of either the exposure or outcome of interest in case-control or cohort study, respectively.⁵⁷

2.6 | Statistical analysis

After descriptive data extraction, supplementary statistical analysis was conducted within studies^{58,59} when appropriate meta-analysis techniques were precluded. In the study by Jordan et al,⁵⁸ pooled mean and standard deviation were calculated from uninjured elite male and female athlete groups to yield a single pooled healthy control group, which was compared to their ACLR group. Similarly, pooled values were calculated from data reported by Kuenze et al,⁵⁹ in ACLR male and female subject groups to create a single ACLR group. Prior to the meta-analysis, pooled averaging was performed to yield a single average value within studies that reported two or more measurement intervals for the early- or late-phase RFD.

Meta-analysis was conducted (R software, Meta version 4.9-2; R Studio, version 3.3.2) to integrate pooled data from (a) studies reporting between-limb RFD asymmetries in individuals post-ACLR and (b) studies comparing RFD between individuals with and without ACLR. Analysis was performed for early-phase RFD < 100 ms (0-30 and 0-50 ms) and late-phase RFD ≥ 100 ms (ie, 0-100, 0-150, 0-200, and 0-250 ms).

The standardized mean differences (SMDs) ($\pm$ 95% confidence intervals—CI) from each trial were calculated and grouped through the use of a random-effect model.⁶⁰ Each

difference of the means was weighted according to the inverse variance method.⁵⁴ SMD outcomes with ≥ 0.2 to < 0.5 were considered small, ≥ 0.5 to < 0.8 were considered medium, and ≥ 0.8 were considered large.^{61,62}

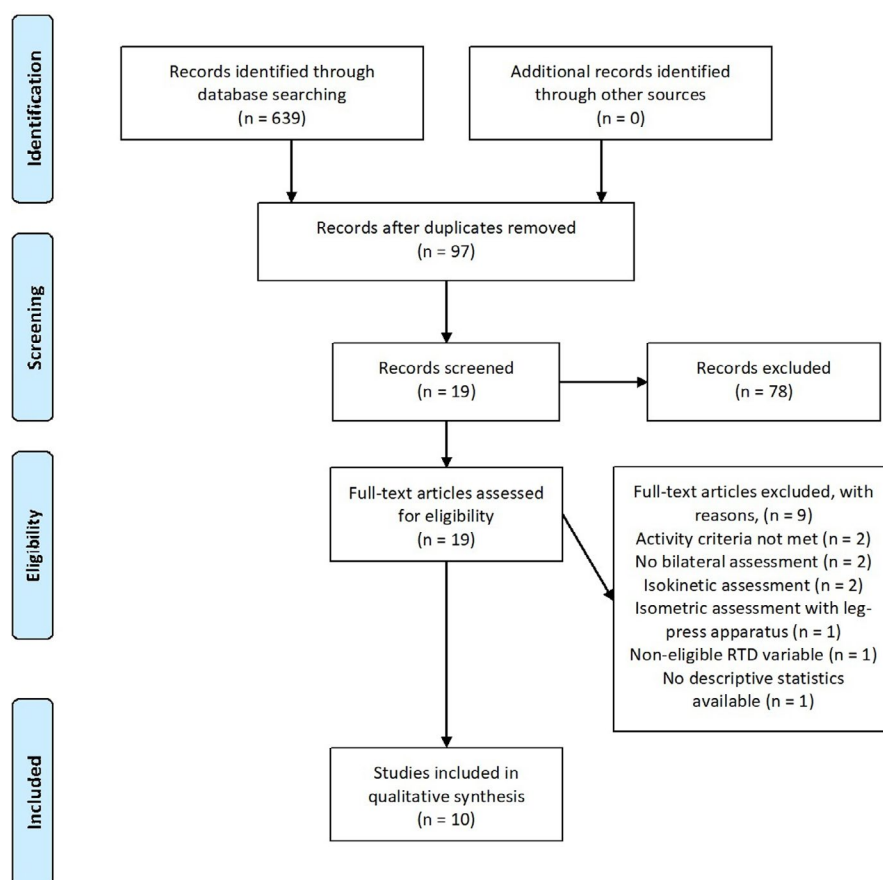
The statistical heterogeneity between the studies was evaluated through I^2 statistics, while between-study variance was assessed using tau-square analysis (τ^2).^{63,64} A scale of magnitude was implemented for the interpretation of statistical heterogeneity of the results for I^2 , where $< 25\%$ was considered as a low magnitude, 50% a medium magnitude, and $> 75\%$ a high magnitude⁶⁴ of statistical heterogeneity. A P value $< .1$ for the chi-square was defined as indicating the presence of heterogeneity, and $\tau^2 > 1$ was defined to suggest the presence of substantial statistical heterogeneity.

3 | RESULTS

3.1 | Study selection

Our database search produced a total of 639 results, where 542 studies were excluded after duplicate removal. After reviewing the remaining 97 abstracts, a total of 19 studies were reviewed in full text, and 10 studies were included for systematic review (flow diagram is depicted in Figure 1).

FIGURE 1 Flowchart of study identification, screening, and inclusion



3.2 | Study characteristics

All studies selected for this systematic review were observational group design studies, and four studies compared ACLR and healthy control groups. Study characteristics are summarized in Table S3. Comprising a total of 246 participants (109 females/137 males), the mean age ranged from 18 to 29.6 years, body mass from 68.18 to 84 kg, and body height from 169 to 181 cm. Duration since ACLR ranged from 4 months up to 49.4 months (Table S3).

3.3 | Risk of bias assessment

From NOS scale assessment, high-quality study was considered to be a study with 9/9 rating. Moderate quality studies were considered to have 7-8/9 rating. None of the included studies were below moderate quality. Four studies^{56,58,65,66} included also an uninjured healthy control group. Healthy uninjured control participants in the four included studies^{56,58,65,66} were matched mostly demographically by sex, age, weight, height, and reported athletic status. Studies showed participant selection bias as studies predominantly included voluntary participants. There were also inconsistencies among four studies^{58,67-69} in reporting associated injuries (eg, meniscal tears, multi-ligament injury, and chondral injury).

3.4 | RFD within-subject between-limb asymmetry after ACLR

3.4.1 | Meta-analysis

In the present analysis, overall pooled SMD for early-phase knee extension RFD (<100 ms) was -1.07 (95% CI: -1.46 , -0.68 ; $I^2 = 22\%$, $\tau^2 = 0.0429$, $P = .28$) (Figure 2), while for late-phase knee extension RFD (≥ 100 ms), SMD was -0.85 (95% CI: -1.27 , -0.42 ; $I^2 = 79\%$, $\tau^2 = 0.3127$, $P = .01$) (Figure 2).

The overall pooled SMD for early-phase knee flexion RFD (<100 ms) was -0.74 (95% CI: -1.19 , -0.29 ; $I^2 = 45\%$, $\tau^2 = 0.1202$, $P = .12$) (Figure 3), while SMD for late-phase knee flexion RFD (≥ 100 ms) was -0.79 (95% CI: -1.19 , -0.39 ; $I^2 = 31\%$, $\tau^2 = 0.0650$, $P = .22$) (Figure 3).

3.4.2 | Descriptive analysis

The within-subject between-limb asymmetry for knee extensor RFD ranged from 7% to 57% and 5% to 32% for knee flexor RFD across the RFD time intervals (ie, 0-250 ms) and post-ACLR time periods (Table S4).

While between-limb asymmetries in knee extensor and flexor RFD assessed from 7 months up to 1 year after ACLR varied between 7% and 55% and 5% and 16%, respectively

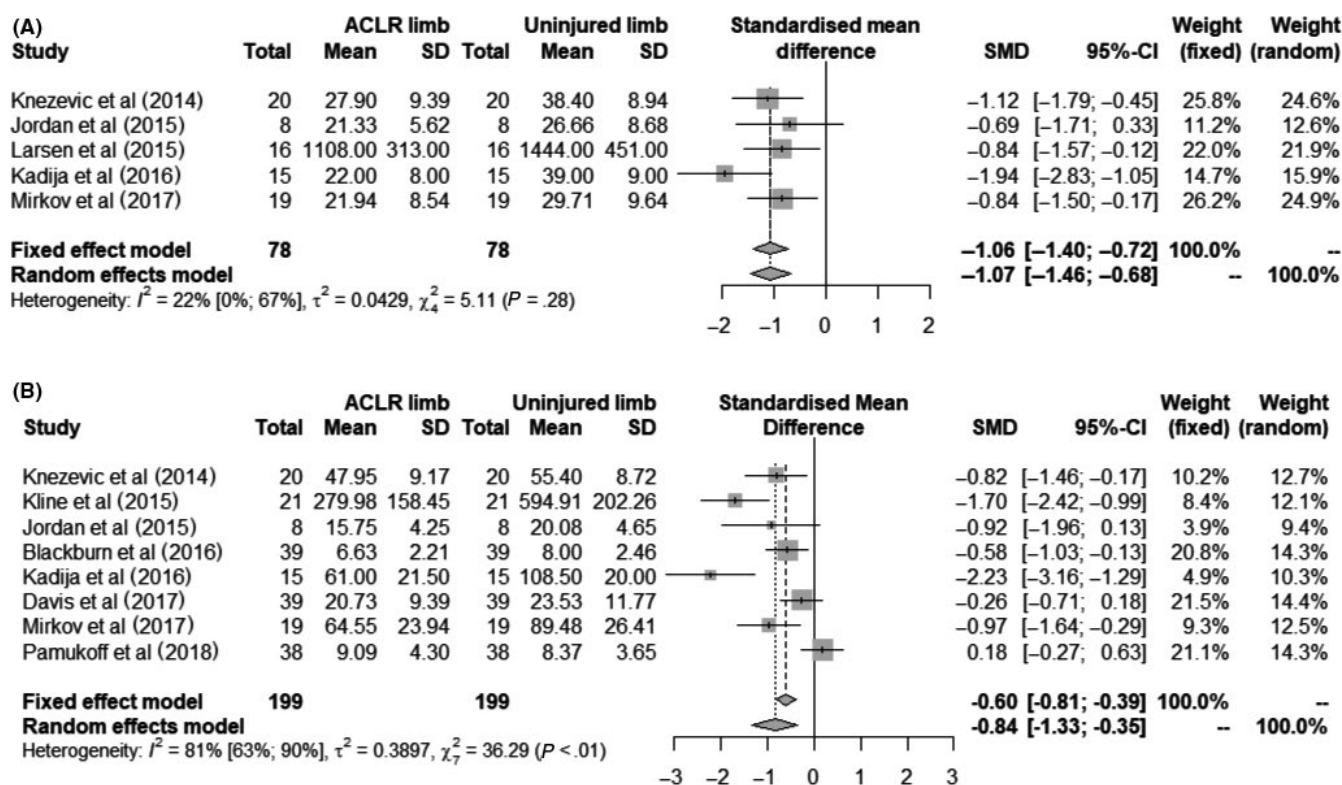


FIGURE 2 ACLR group between-limb asymmetry in knee extensor RFD. ACLR group between-limb asymmetry in knee extensor RFD based on (A) early-phase RFD (<100 ms) and (B) late-phase RFD (≥ 100 ms)

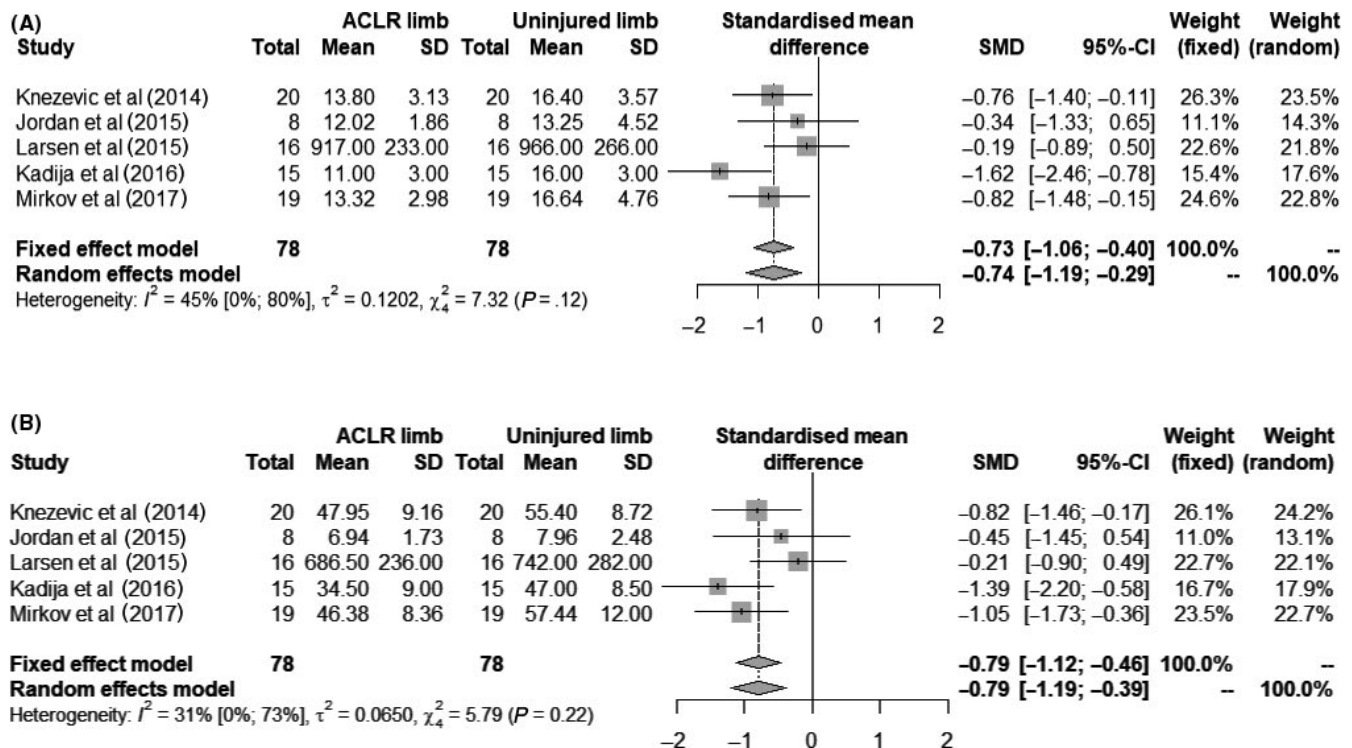


FIGURE 3 ACLR group between-limb asymmetry in knee flexor RFD. ACLR group between-limb asymmetry in knee flexor RFD assessed as (A) early-phase RFD (<100 ms) and (B) late-phase RFD (≥ 100 ms)

(RFD time intervals between 0 and 250 ms), statistically significant ($P < .05$) between-limb asymmetries were found to exist only for knee extensor RFD (Table S4).

Between-limb asymmetry for knee extensor and flexor RFD assessed 2 years after ACLR ranged between 20% and 23% and 9% and 13%, respectively (RFD time intervals 0-250 ms)⁵² (Table S4). In addition, a 11%-22% range in knee extensor RFD asymmetry (0-100 ms and 0-200 ms) was reported 4 years after ACLR.⁶⁸ Interestingly, another study in individuals examined 4 years after ACLR reported negligible between-limb asymmetry in RFD for the knee extensors except for (0-100 ms) RFD when compared to healthy controls (SMD > 0.5)⁶⁶ (Table S4).

3.5 | ACLR vs healthy controls

3.5.1 | Meta-analysis

Overall pooled SMD for early-phase knee extension RFD (0-100 ms) was -1.42 (95% CI: -2.10 , -0.73 ; $I^2 = 52\%$, $\tau^2 = 0.1883$, $P = .13$) (Figure S1), whereas for late-phase knee extension RFD (≥ 100 ms), SMD was -1.09 (95% CI: -1.81 , -0.38 ; $I^2 = 76\%$, $\tau^2 = 0.3928$, $P = .01$) (Figure S1).

Overall pooled SMD for early-phase knee flexion (<100 ms) was -0.78 (95% CI: -1.96 , -0.39 ; $I^2 = 85\%$, $\tau^2 = 0.9175$, $P = .01$) (Figure S2), while SMD for late-phase

knee flexion RFD (≥ 100 ms) was -1.14 (95% CI: -1.60 , -0.67 ; $I^2 = 0\%$, $\tau^2 = 0$, $P = .53$) (Figure S2).

3.5.2 | Descriptive analysis

Substantial differences in RFD were found when comparing the injured limb in individuals post-ACLR to healthy controls (Table S5).^{56,58,65,66} The largest RFD deficits were noted 4 months⁵⁶ after ACLR for the knee extensors (14%-72% reduced) and knee flexors (2%-59% reduced) when comparing individuals post-ACLR with healthy controls (RFD time intervals between 0 and 200 ms) (Table S5). Differences in RFD were also noted 1⁶⁵ to 2 years⁵⁸ after ACLR for the knee extensors (8%-39% reduced) and knee flexors (3%-20% reduced) when comparing the injured limb of individuals post-ACLR to the uninjured limbs of healthy controls. Finally, reductions of 5%-23% in RFD could be noted as much as 4 years⁶⁶ post-ACLR when compared to healthy controls (Table S5).

4 | DISCUSSION

The primary aim of this systematic review and meta-analysis was to examine whether knee extensor and flexor RFD deficits existed for the ACLR limb compared to the contralateral uninjured limb and to the limbs of individuals without

ACLR. Knee extensor RFD data comparing the ACLR limb to the uninjured limb were pooled from 10 studies comprising a total of 246 participants^{35,47,49,50,57,59,65,66,70,71} and five studies totaling 78 participants for knee flexor RFD.^{56,58,65,71,72} Comparisons between participants with and without ACLR were pooled from four studies ($n = 211$ participants).^{47,66,69,73}

As the main study finding, ACLR resulted in substantial between-limb asymmetry (ie, ACLR limb deficits) in RFD, evidenced by moderate-to-large (>0.7) SMDs in early- and late-phase RFD for both the knee extensors and flexors. Large (>1.0) SMDs were also observed when comparing individuals with ACLR to uninjured controls for early- and late-phase RFD for both the knee extensors and flexors. Further, the present meta-analysis revealed moderate-to-large statistical heterogeneity for between-limb comparisons in ACLR participants (injured limb vs uninjured contralateral limb) and for comparisons between ACLR individuals and uninjured healthy controls.

The present systematic review suggests that between-limb asymmetries in RFD after ACLR may persist for several years post-operatively for both the knee extensors and the knee flexors. Between-limb asymmetry in the knee extensor RFD (26%-57%) was highest four to 7 months post-ACLR^{56,59,69,71,72} and decreased through to 1 year after ACLR⁶⁵ (knee extensor RFD asymmetry 21%-29%). Knee extensor RFD asymmetry was still present 2⁵⁸ (20%-23%) to 4 years^{63,67} (11%-22%) after ACLR. Interestingly, participants in a study by Pamukoff et al⁶⁶ demonstrated full symmetry of knee extensors 4 years after ACLR and RFD deficits were observed only when comparing the ACLR group to an uninjured control group (5%-23%). Between-limb asymmetries in knee flexor RFD were of a moderate-to-high magnitude four to 6 months after ACLR (11%-32%)^{71,72} while approaching small-to-moderate magnitude at later time points of 1⁶⁵ to 2⁵⁸ years after ACLR surgery (5%-13%).

4.1 | Influence of ACLR graft type and knee flexion angle on RFD between-limb asymmetry

It was not possible to statistically evaluate the influence of ACLR graft type on RFD between studies. Bone-patellar tendon-bone (BPTB) autografts were mostly used for ACLR in the present analysis.^{56,58,59,65-68,71-73} This may explain the more marked and persistent between-limb asymmetry in knee extensor RFD compared to knee flexor RFD.^{9,71} In support of this notion, between-limb asymmetry deficits based on MVC also have been observed for the knee extensors in individuals with BPTB autograft.^{14,20} In terms of knee flexor (hamstring muscle) function, while the biceps femoris muscle may contribute to RFD strength assessments at 15-30° of knee flexion,^{77,78} the semitendinosus and gracilis muscles are strong agonists for knee flexion

torque beyond 75° of knee flexion.^{58,73} As the semitendinosus autograft is a common ACLR technique, this highlights the importance of considering the torque-joint angle relationship when assessing RFD capacity. Jordan et al⁵⁸ study was the only study to assess knee flexion at larger angles (70°). It is challenging to interpret the possible influence of the semitendinosus autograft technique on knee flexor RFD. This remains an important question that should be given specific attention in future studies evaluating RFD capacity after ACLR.

4.2 | ACLR and neuromuscular-related factors affecting RFD between-limb asymmetry

Since RFD is correlated with maximal muscle strength,⁴⁰ mechanisms underpinning between-limb asymmetries in maximal muscle strength (MVC) after ACLR may also underpin the observed RFD deficiencies. Changes in the activation of cortical brain areas^{74,79} and reduced excitability in descending corticospinal^{76,80-83} and spinal reflex pathways^{81,83-85} have been observed after ACL injury, contributing to a persistent depression in maximal knee extensor strength.⁸⁰ More specifically, increased hamstring co-activation,⁸⁶ gamma-loop dysfunction,¹⁷ and AMI¹⁶ might explain the observed between-limb asymmetries and RFD deficits after ACLR. Pamukoff et al⁸⁶ described increased hamstring co-activation in ACLR knees (50.7 ± 21.3 months after ACLR) compared to healthy uninjured controls. Interestingly, increased hamstring co-activation was associated with reductions in knee extensor peak torque ($r = -.39$), Peak RTD ($r = -.30$), central activation ratio (CAR) ($r = -.40$), and electromyography (EMG) amplitude ($r = -.30$).⁸⁶ Additionally, the ACL is known to have sensory functions.^{17,87} Loss of afferent feedback from ACL rupture itself and surgical reconstruction may lead to gamma-loop dysfunction and impaired Ia afferent input to spinal motoneurons during the first 6 months after ACLR.¹⁷ This important issue is also believed to result in an inability to fully activate fast-twitch motor units.¹⁷ Konishi et al^{17,88,89} suggested that gamma-loop dysfunction could also diminish the contralateral limb function after ACL injury, although these findings have been questioned.¹⁰ Regarding AMI (also referred to impaired voluntary activation⁵⁶), it is thought that knee joint damage causes alterations in afferent CNS input from efferent input of knee joint receptors, resulting in ongoing reflex inhibition that leads to an inability to activate high-threshold motor units during voluntary quadriceps activation.^{15,16} AMI is multifactorial and influenced by joint inflammation, pain, structural damage, joint swelling, and elevated joint laxity.¹⁶ More importantly, it should be noted that AMI influences both limbs after unilateral ACLR and may blunt adaptations evoked by intensive resistance training.^{14,15}

4.3 | ACLR and intrinsic muscular factors affecting RFD between-limb asymmetry

Studies have described reduced knee extensor muscle volume and cross-sectional area following ACLR compared to the uninjured side⁹⁰ and to healthy controls.⁹¹ Similar findings have been described for the knee flexor muscles when compared to the uninjured side⁹²⁻⁹⁸ and uninjured controls.⁹² Notably, more profound reductions in semitendinosus muscle CSA and volume compared to the biceps femoris and semimembranosus muscles have been reported following semitendinosus autograft, commonly used in ACL reconstruction surgery.^{97,99-101} Also, Messer et al¹⁰⁰ showed persistent semitendinosus activation deficiency 1-6 years after ACLR during eccentric knee extensor exercise. Such persistent activation deficiency of the semitendinosus muscle could explain the long-term semitendinosus atrophy observed after ACLR.^{93,97,98,100}

Noehren et al¹⁰² studied cellular and morphological changes in the vastus lateralis muscle before and after ACLR. A reduced muscle volume and CSA were observed alongside significant reductions in type IIa muscle fiber CSA, muscle fascicle pennation angle, and muscle stem cells (satellite cells) per fiber, accompanied by increased extracellular matrix volume.¹⁰² Following surgery, alterations in the injured limb persisted, characterized by selective reductions in type IIa fibers and increased abundance of type IIa/x hybrid fibers.¹⁰² Increased extracellular matrix content may reduce force generation capability due to less area occupied by active contractile filaments.¹⁰² Also, satellite cells serve a critical role in muscle regeneration and repair^{103,104} and may also play an important role in the adaptive potential of muscle throughout ACLR rehabilitation.¹⁰² In support of this notion, reduced satellite cell density in the knee extensor (vastus lateralis) has been reported 59 ± 67 days after ACL injury.¹⁰⁵

Studies describing intrinsic changes to the muscle fibers after ACL injury and subsequent ACLR generally are lacking.⁷⁵ Mendias et al¹⁰⁶ reported elevated myocellular levels of atrophy-inducing cytokines (myostatin and TGF- β) along with substantially reduced knee extensor strength compared to the uninjured side 5 weeks after ACLR. Gumucio et al⁷⁵ obtained vastus lateralis muscle biopsies 1, 2, and 6 months after ACLR, which revealed that individual myofibers had impaired intrinsic force generation capacity as reflected by reduced specific tension (force/CSA). Given that muscle volume and cross-sectional area, and muscle fascicle pennation angle influence maximal muscle force generation capacity,^{18,19} it is likely that these morphological changes contribute to the RFD deficits described here in the knee extensor⁹⁰ and knee flexor^{82,88,89,94,96-98} muscles after ACLR.

It may be difficult to establish the specific contribution of neural vs peripheral muscular factors, on RFD capacity after ACLR based on traditional RFD analysis alone. However,

the influence of neural and muscular parameters on contractile RFD may to some extent be inferred from early-phase RFD analysis (<100 ms)¹⁰⁷ or by employing sequential (stepwise) RFD analysis in an already activated muscle (eg, 100-150 ms).³⁹ In the present analysis, the influence of ACLR on neural and peripheral intrinsic muscle contractile determinants of RFD was difficult to pinpoint given that no studies reported such sequential RFD data, and more direct muscle biopsy data and/or EMG data or equivalent measures of neuromuscular activity also were missing. Nevertheless, our analysis showed persistent between-limb asymmetry data based on early-phase RFD recordings (<100 ms), suggesting neurally mediated between-limb asymmetries may be present after ACLR, since motor unit recruitment and motor neuron discharge rate are considered the strongest determinants of early-phase RFD.^{39,40}

Mirkov et al⁵⁶ reported substantial RFD deficits in individuals post-ACLR compared to healthy uninjured controls 4 months after ACLR for both knee extensor RFD (0-180 ms) and knee flexor RFD (0-150 ms) including reduced RFD for the ACLR limb of 42% and 32%, respectively. Further, no between-group differences were found for knee extensor or flexor MVC peak torque. Underpinning mechanisms of the divergent recovery of RFD and MVC abilities were unclear, but as the rehabilitation programs were identical for all the ACLR study participants, the persistent time-based RFD deficits might be explained by pre-surgical gamma-loop dysfunction, AMI, proprioception deficits, or involvement of spinal and supraspinal inhibitory mechanisms.⁵⁶ Studies by Larsen et al⁶⁵ and Jordan et al⁵⁸ have similarly confirmed influence of ACLR on RFD. A small-to-moderate relative RFD difference (8%-17%) between the ACLR group and the healthy control group was noted by Larsen et al⁶⁵. Likewise, in a sample of elite alpine ski racers, Jordan et al⁵⁸ reported small-to-moderate relative differences in RFD (3%-18%) when comparing the ACLR group to a matched group of uninjured controls. The participant activity level may have contributed to these discrepant findings as Larsen et al⁶⁵ examined recreationally active individuals whereas Jordan et al⁵⁸ included elite-level alpine ski racers, who were presumably highly trained. As a 2-year post-surgical recovery period is expected after ACLR to restore normal/previous knee joint health and function,¹⁰⁸ the findings reported by Larsen et al⁶⁵ and Jordan et al⁵⁸ may be related to post-operative time periods of insufficient duration along with factors such as AMI,¹⁶ chronic elevations in hamstring co-contraction,⁸⁶ and persistent atrophy of the knee extensors^{82,91,94,109-111} and knee flexors^{82,88,89,94,96-98,100} after ACLR.

Considering the potential influence of bilateral AMI after unilateral ACLR,^{14,15} the goal of restoring RFD ability to the level of the uninjured contralateral limb using a LSI may underestimate the true recovery from ACLR surgery.⁵⁶ Thus, pre-injury testing,¹¹² pre-surgery testing,^{113,114} comparison

to matched uninjured controls,^{56,58,65} and/or a comparison to normative values³³ should be considered alongside a LSI when evaluating RFD and maximal strength capacity (MVC) after ACLR.

4.4 | Changes in RFD with rehabilitation after ACLR

Rehabilitation protocols were informed briefly and not summarized in fine detail in the studies included in the present analysis.^{56,58,59,65-68,71-73} It is possible, therefore, that insufficient rehabilitation efforts are at least partially responsible for the persistent RFD between-limb asymmetries observed after ACLR. Maximal and explosive-type (ie, performed with high RFD) resistance training^{41,107,115} including ballistic training (ie, the use of a relatively light external load with maximal intentional velocity), ensuring maximal intent to generate rapid force, and sufficient similarity between training and testing movement patterns are all factors known to enhance RFD.^{116,117} High-intensity training protocols may not be typical components of resistance training programs employed after ACLR, at least not until the late rehabilitation phase after knee mobility/function has been fully restored in order to ensure patient safety and to maximize tissue healing.¹¹⁷ Nevertheless, designing resistance training exercise regimes to increase RFD should be an integral component of post-surgical ACLR rehabilitation especially in athletes where the goal is to restore neural drive and rapid force capacity (RFD and RTD) before entering into field-based rehabilitation and RTS.¹¹⁷

4.5 | Modifiable neuromuscular and physiological risk factors

Studies aimed at identifying neuromuscular injury risk factors¹¹⁸ are likely important for reducing ACL injury risk. Injury risk factor identification from athlete pre-season isokinetic strength assessments seems to play only a minor role in the prevention of ACL injury¹¹⁹ or hamstring strain.¹²⁰ Thus, future studies incorporating an assessment of rapid force generating ability such as RFD, RFD H/Q³³ ratio, the influence of performance fatigability on neuromuscular function such as hamstring-to-quadriceps fatigue ratio¹²¹ and acute match-induced decrements in hamstring vs quadriceps RFD^{53,122} may prove useful for ACL injury prevention efforts when considering potential, modifiable neuromuscular risk factors. However, due to the technical nature of RFD assessments (eg, equipment, methodological issues, and analysis technique), RFD testing may be unavailable in conventional rehabilitation settings.^{39,40} Nevertheless, an understanding

of the physiological determinants of RFD, the knowledge of persistent RFD vs MVC knee flexor/extensor deficits, and application of RFD resistance training methods remain important throughout ACLR rehabilitation.¹¹⁷

Return to sport assessment should also be multifactorial and preferably involve more than a single outcome variable.^{23,123} RTS assessments may further require customization to the performance level of the individual¹²⁴ as there are reports that 83% of elite ACLR athletes returned to their pre-injury performance level.¹²⁵ Thus, physical/physiological performance factors, which based on the present review include RFD, may explain the better rate of return to sport of elite athletes compared to recreational athletes. While modifiable neuromuscular factors such as quadriceps strength seem to have a supporting³⁵ and weak association^{26,126} with a successful RTS after ACLR, the present data suggest the relevance of RFD testing to identify trainable neuromuscular deficits in preparation for RTS and may be especially important for athletes who are required to perform rapid/high-force movements in training and competition.

4.6 | Study limitations

More research appears necessary to confirm the findings obtained from this systematic review. In particular, long-term follow-up studies should be conducted to describe more precisely the time and rehabilitation-dependent changes in RFD after ACLR. Perhaps more importantly, specific rehabilitation and resistance training protocols should be described in greater detail, especially in studies addressing between-limb asymmetries in RFD or differences herein compared to healthy controls. Also, in the literature there is lack of data on between-limb RFD symmetry prior to ACLR, which limits an exploration of the impact of ACLR on RFD capacity and does not permit to fully explore magnitude effect of ACLR on RFD capacity. Interestingly, high pre-operative quadriceps strength LSI (>80%) has been positively associated with accelerated quadriceps LSI recovery when compared to lower pre-operative LSI.¹¹³ ACLR surgical procedures including the use of autograft or allograft techniques and their effect on knee extensor and flexor RFD should also be examined. Of importance is evaluating the co-morbidity effects from the use of autografts (eg, bone-patellar tendon-bone or semitendinosus autograft) compared to allografts. Further, as the semitendinosus is particularly important for generating knee flexor torque at large (flexed) angles of knee flexion, knee flexor RFD and MVC assessments obtained at 70° of knee flexion or below seem critical.⁷⁷

Use of activity-level inclusion criteria is recommended since only four previous studies^{59,66,68,69} have provided data

on the Tegner activity score.⁵⁵ Three studies included male athletes⁷¹ or recreational athletes^{65,67} without further information on athletic status, and three studies comprised competitive-level athletes recruited from various sports.^{56,58,72} Jordan et al⁵⁸ examined elite athletes who were compared to matched controls (National alpine ski racers), Mirkov et al⁵⁶ involved amateur athletes, and Larsen et al⁶⁵ recruited matched healthy control participants based on activity level using estimates of occupational and spare-time physical activity. Kline et al⁶⁹ reported the largest RFD between-limb asymmetries (51%-57%), which were suggested to differ from reports by Knezevic et al⁷² (between-limb RFD asymmetries of 27%-28%) due to the activity status of the experimental subjects (recreational athletes⁶⁹ vs competitive athletes⁷²). Thus, the specific activity levels of study participants should be considered when comparing and interpreting results between different studies.

Confounding covariate modifiers across studies may have caused distortion between the exposure to ACLR and the resulting change in RFD. Possible confounding covariate modifiers between studies include the influence of ACLR graft selection, associated injuries with ACL tears (meniscal tears, chondral injury, and multi-ligament injury), time from ACLR to the RFD testing time point, participant athletic status, and the potential for variability in rehabilitation and retraining techniques. Covariate modifiers may have also increased the statistical heterogeneity in the present meta-analysis. Due to small number of included studies, explanatory analysis such as subgroup analysis or meta-regression was not feasible to conduct. Finally, the present results must be interpreted with caution due to the relatively small number of subjects ($n = 246$) included in the meta-analysis, which comprised a dominant proportion of male participants and a relatively young age of the study participants.

5 | CONCLUSIONS

Based on the present meta-analysis, significant deficits in rapid force capacity (rate of force development; RFD) for the knee extensors and flexors persist for affected limb of individuals with ACLR. Consequently, RFD assessments should be considered throughout the ACLR rehabilitation phase prior to RTS in order to monitor potential training-related neuromuscular deficits. RFD testing may identify deficits in mechanical lower limb muscle function not revealed with MVC testing alone and could be used to indirectly monitor neural and muscular changes occurring consequent to rehabilitation and reconditioning after ACLR. It is recommended that training methods to increase RFD be considered after ACLR prior to the RTS transition and commencing high-force/rapid sport-specific movements.

6 | PERSPECTIVES

6.1 | Findings

Substantial deficits in rapid force capacity (rate of force development: RFD) manifested as significant between-limb asymmetries were observed after ACLR, which persisted up to 4 years after ACLR. Likewise, persistent RFD deficiencies after ACLR also were apparent when affected limbs of individuals post-ACLR were compared to healthy uninjured controls.

6.2 | Implications

Between-limb asymmetries in RFD should be evaluated if possible alongside MVC assessment in the consideration of RTS following ACLR. Rehabilitation strategies to decrease RFD asymmetries of the quadriceps and hamstring muscles should be implemented throughout the time course of rehabilitation before considering on-field-based rehabilitation activities.

CONFLICT OF INTEREST

None to declare.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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